

TD 3 - Financial Econometrics

January 2026, Arnaud Capitan

Part I : Capitalization-weighted portfolio

We assume that the number of assets considered in the exercise is $m \in \mathbb{N}^*$. The capitalization-weighted portfolio (CWP) is constructed such that :

$$\forall i \in [1, m], \quad w_{i,t} = \frac{v_{i,t}}{v_t} = \frac{n_{i,t}p_{i,t}}{\sum_{1 \leq j \leq m} n_{j,t}p_{j,t}} = \frac{C_{i,t}}{C_t}$$

$p_{i,t}$, $n_{i,t}$, $v_{i,t}$ respectfully denote the price, the number of shares, and the value of asset i . $C_{i,t}$ is the capitalization of asset i .

It is also assumed that the number of each asset in the market is fixed : $C_{i,t} = N_i p_{i,t}$

Question 1 :

From this question we get an extra assumption, that the composition of our portfolio is fixed, $\forall i \in [1, m]$, $n_{i,t} = n_t$

$$\text{CWP at time } t = 0 : \quad \forall i \in [1, m], \quad w_{i,0} = \frac{n_i p_{i,0}}{\sum_{1 \leq j \leq m} n_j p_{j,0}} = \frac{N_i p_{i,0}}{\sum_{1 \leq j \leq m} N_j p_{j,0}}$$

Assuming $p_{i,0} \neq 0 \quad \forall i \in [1, m]$, and that the global portfolio capitalization verifies $C_t > 0$, we get :

$$\forall i \in [1, m], \quad n_i = \frac{\sum_{1 \leq j \leq m} n_j p_{j,0}}{\sum_{1 \leq j \leq m} N_j p_{j,0}} N_i \quad (1)$$

At time $t > 0$, we get, $\forall i \in [1, m]$:

$$w_{i,t} = \frac{n_i p_{i,t}}{\sum_{1 \leq j \leq m} n_j p_{j,t}}$$

Using expression (1), we get :

$$w_{i,t} = \frac{\frac{\sum_{1 \leq j \leq m} n_j p_{j,0}}{\sum_{1 \leq j \leq m} N_j p_{j,0}} N_i p_{i,t}}{\sum_{1 \leq j \leq m} \frac{\sum_{1 \leq k \leq m} n_k p_{k,0}}{\sum_{1 \leq k \leq m} N_k p_{k,0}} N_j p_{j,t}}$$

$$w_{i,t} = \frac{\frac{\sum_{1 \leq j \leq m} n_j p_{j,0}}{\sum_{1 \leq j \leq m} N_j p_{j,0}} N_i p_{i,t}}{\frac{\sum_{1 \leq k \leq m} n_k p_{k,0}}{\sum_{1 \leq k \leq m} N_k p_{k,0}} \sum_{1 \leq j \leq m} N_j p_{j,t}}$$

$$w_{i,t} = \frac{N_i p_{i,t}}{\sum_{1 \leq j \leq m} N_j p_{j,t}}$$

$$w_{i,t} = \frac{C_{i,t}}{C_t}$$

So a CWP at time $t = 0$ whose composition is fixed will remain a CWP at time $t > 0$.

Question 2 :

Since we have $\forall i \in [1, m], \quad v_{i,t} = v_{i,0} \frac{p_{i,t}}{p_{i,0}}$

$$v_t = \sum_{1 \leq i \leq m} v_{i,t} \quad \forall t \geq 0$$

$$v_t = \sum_{1 \leq i \leq m} v_{i,0} \frac{p_{i,t}}{p_{i,0}}$$

Since by definition, we have $w_{i,t} = \frac{v_{i,t}}{v_t} \quad \forall t$, we use this equality at $t = 0$ to get :

$$v_{i,0} = v_0 w_{i_0}$$

Thus, we obtain :

$$v_t = \sum_{1 \leq i \leq m} v_0 w_{i_0} \frac{p_{i,t}}{p_{i,0}}$$

$$v_t = v_0 \sum_{1 \leq i \leq m} \frac{p_{i,t}}{p_{i,0}} w_{i_0}$$

Question 3 :

We have $m = 3$, and we assume that we keep the same assumption as in question 1, $n_{i,t} = n_i$

We have :

- The prices of each asset at all time $p_{i,t}$
- The initial value v_0

We just need to compute the initial weight $w_{i,0} = \frac{C_{i,0}}{C_0}$

Since we obtained this formula from question 2 :

$$v_t = v_0 \sum_{1 \leq i \leq m} \frac{p_{i,t}}{p_{i,0}} w_{i,0}$$

We can obtain the value at all time of our CWP, and compute the returns from the value/price of our CWP.

$$r_t = \frac{v_t}{v_{t-1}} - 1$$

I will also add a column to give the excess returns of our portfolio by subtracting the risk free returns to the previously computed returns of the CWP.

Question 4 :

To test the efficiency of this portfolio on October 29 2021, we verify that the composition of our portfolio is proportional to the composition of the efficient portfolio.

To compute the efficient portfolio :

- We first estimate the net daily returns of each asset $r_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}} - 1$
- We then estimate the expected return, volatility, then annualized return and annualized volatility
- We compute the excess returns $\tilde{\mu}$ for each firm, by subtracting the risk free returns
- We compute the covariance-variance matrix using the correlation matrix and the annualized variance, since :

$$\hat{\gamma}_{i,j} = \rho_{i,j} \sqrt{\hat{\gamma}_{i,i} \hat{\gamma}_{j,j}}$$

- Then we inverse the obtained covariance matrix Ω
- Finally we obtain the efficient portfolio :

$$w^* = \Omega^{-1} \tilde{\mu}$$

We then compare this portfolio with the CWP :

$$w_{i,t} = \frac{C_{i,t}}{C_t}$$

We use the hypothesis H_0 :

If the CWP is efficient, then $\exists k \in \mathbb{R}$ such that $w^* = kw_t$

The efficient portfolio reads :

$$w_1^* = -0.32 \quad w_2^* = 0.32 \quad w_3^* = 0.99$$

Which means that the efficient portfolio is shorting the firm Ford Motor, and going long both Johnson n Johnson and JPMorgan Chase. It also borrows money at the risk free rate in order to invest more.

To compute the CWP weights :

$$C_{i,t} = N_i p_{i,t}$$

$$C_{i,t} = C_{i,0} \frac{p_{i,t}}{p_{i,0}}$$

Thus :

$$w_{i,t} = \frac{C_{i,t}}{\sum C_{i,t}}$$

The CWP portfolio reads :

$$w_1^* = 0.061 \quad w_2^* = 0.414 \quad w_3^* = 0.524$$

It seems obvious that the CWP is not efficient. It can be explained by the fact that a CWP portfolio cannot hold short positions, while the efficient portfolio seems to use this ability to short Ford and add a long position on JPMorgan.

Question 5 :

We want to estimate the 3-factor Fama-French model on the CWP portfolio net return series.

We are given the data to estimate the model : Market portfolio excess returns Mkt - RF, SMB returns, and HML returns

We will use as input the excess returns of the CWP to make the regression of the factors :

$$r_{CWP-RF,t} = \alpha + \beta_1 r_{MKT-RF,t} + \beta_2 r_{SMB,t} + \beta_3 r_{HML,t}$$

We get the following coefficients :

- $\beta_3 = 0.4388$ HML coefficient
- $\beta_2 = -0.2910$ SMB coefficient
- $\beta_1 = 1.0957$ Market-RF coefficient
- $\alpha = -0.0004$ α coefficient (close to 0, which is a good indicator that the regression worked)

We wish to compare the 1-factor model (market) against the 3-factor model.

This is equivalent to verifying the hypothesis :

$$H_0 : \beta_3 = \beta_2 = 0$$

To do this verification, we will do a Wald statistic test :

Assuming that β_2 and β_3 are uncorrelated, under H_0 we have that :

$$W = \frac{\beta_2^2}{\sigma_2^2} + \frac{\beta_3^2}{\sigma_3^2} \sim \chi^2(2)$$

If W were to follow such distribution, we would have :

$$\mathbb{P}(W \leq 5.99) = 0.95$$

But we get that $W = 151.92$, which is much higher than the 95% quantile, which means that we reject H_0 .

The results of this statistic test is that the 3-factor model does in fact give more information, and that the parameters β_2 and β_3 , respectively the SMB and HML coefficients, are significant.

Part 2 : Efficient portfolio and investment views

Question 1 :

This result has already been obtained during question 4.

$$w^* = \Omega^{-1} \tilde{\mu}$$

$$\bar{w}^* = \frac{\Omega^{-1} \tilde{\mu}}{e^T \Omega^{-1} \tilde{\mu}}$$

We get :

$$\bar{w}^* = [-0.320, 0.329, 0.991]^T$$

This means that the efficient portfolio is holding a short position on Ford to finance the long positions of Johnson n Johnson and JPMorgan.

Question 2 :

The risk aversion coefficient associated to this portfolio is :

$$\bar{\lambda} = e^T \Omega^{-1} \tilde{\mu} = 2.545$$

This means that this portfolio is more conservative ($\bar{\lambda} > 1$) while still holding some risks. It seems realistic.

Question 3 :

The coefficient $\tau = \frac{1}{n}$ comes from the statistical estimator of the expected net excess return.

In fact, denoting $\mathbb{E}_t[z_{t+1}] = \pi$, the empirical excess returns verify by the Central Limit Theorem :

$$\sqrt{n}(\tilde{\mu}_n - \pi) \sim \mathcal{N}(0, \Omega)$$

Thus obtaining :

$$\tilde{\mu}_n \sim \mathcal{N}(\pi, \frac{\Omega}{n})$$

And we get $\tau = \frac{1}{n}$ with n being the sample size.

Question 4 :

To compute π such that $w^C = w^{(m)}$, we first compute $w^{(m)}$ as a function of π .

$$w^{(m)} = \underset{w}{\operatorname{argmax}} \quad w^T \pi - \frac{\bar{\lambda}}{2} w^T \Omega w$$

First order condition :

$$\pi - \bar{\lambda} \Omega w^{(m)} = 0$$

Since we want $w^C = w^{(m)}$, π must verify :

$$\pi = \bar{\lambda} \Omega w^C$$

Which is equivalent to :

$$w^C = \frac{1}{\bar{\lambda}} \Omega^{-1} \pi$$

This expression looks the same as :

$$w^* = \frac{1}{\bar{\lambda}} \Omega^{-1} \tilde{\mu}$$

This means that we are looking for the value π which is the expected equilibrium returns that makes the capitalization-weighted portfolio an efficient portfolio.

We get the value : $\pi = [11.28\%, 7.23\%, 14.29\%]^T$

We had obtained previously that the annualized excess returns of each asset were respectively :

$$\tilde{\mu}_n = [6.78\%, 7.51\%, 17.95\%]^T$$

Thus proving that the CWP portfolio is in fact not an efficient portfolio once again, as the annualized realized excess returns are not close (except for Johnson n Johnson) to π which would make the CWP efficient.

Question 5 :

$$V = P\tilde{\mu} + \varepsilon$$

Where $\varepsilon \sim \mathcal{N}(0, \Sigma)$, independent from $\tilde{\mu}$.

From the Black-Litterman model, we have that $\tilde{\mu} \sim \mathcal{N}(\pi, \tau\Omega)$, independent of ε .

First, we want to determine the distribution of V .

Since V is a linear combination of multivariate gaussian vectors, V is also a gaussian vector, $V \sim \mathcal{N}(\mu_V, \Sigma_V)$

We now aim to find the values (μ_V, Σ_V) :

$$\mu_V = \mathbb{E}[V] = P\mathbb{E}[\tilde{\mu}] = P\pi$$

$$\Sigma_V = Var(V) = PVar(\tilde{\mu})P^T + Var(\varepsilon)$$

$$\Sigma_V = \tau P\Omega P^T + \Sigma$$

$$\text{Thus, } V \sim \mathcal{N}(P\pi, \tau P\Omega P^T + \Sigma)$$

Second, we want to determine the conditional distribution of V given $\tilde{\mu}$.

Since V is still a linear combination of multivariate gaussian vectors, V is also a gaussian vector, thus :

$$V|\tilde{\mu} \sim \mathcal{N}(\mu_{V|\tilde{\mu}}, \Sigma_{V|\tilde{\mu}})$$

We now compute $(\mu_{V|\tilde{\mu}}, \Sigma_{V|\tilde{\mu}})$:

$$\mu_{V|\tilde{\mu}} = \mathbb{E}[V|\tilde{\mu}] = P\tilde{\mu}$$

$$\Sigma_{V|\tilde{\mu}} = Var(V|\tilde{\mu}) = \Sigma$$

Thus we have :

$$V|\tilde{\mu} \sim \mathcal{N}(P\tilde{\mu}, \Sigma)$$

Question 6 :

We want to show that $z_t|V = v$ is Gaussian.

We have that :

$$\begin{aligned} z_t|\tilde{\mu} &\sim \mathcal{N}(\tilde{\mu}, \Omega) \\ \tilde{\mu} &\sim \mathcal{N}(\pi, \tau\Omega) \\ V|\tilde{\mu} &\sim \mathcal{N}(P\tilde{\mu}, \Sigma) \end{aligned}$$

We can write that :

$$z_t = \tilde{\mu} + \eta_t$$

With $\eta_t \sim \mathcal{N}(0, \Omega)$ being independent of $\tilde{\mu}$ and ε .

We have that :

$$\begin{bmatrix} z_t \\ V \end{bmatrix} = \begin{bmatrix} I & I & 0 \\ P & 0 & I \end{bmatrix} \begin{bmatrix} \tilde{\mu} \\ \eta_t \\ \varepsilon \end{bmatrix}$$

Since we have a linear combination of independent Gaussian variable, (z_t, V) is also Gaussian.

Therefore $z_t|V = v$ is also Gaussian because of their independence.

Question 7 :

For $\tau = \frac{1}{n}$ and $c = 10$, we get that :

$$\tilde{\mu}^{BL} = [11.294\%, 7.157\%, 14.230\%]^T$$

$$\Omega^{BL} = \begin{bmatrix} 0.111 & 0.018 & 0.057 \\ 0.018 & 0.036 & 0.024 \\ 0.057 & 0.024 & 0.082 \end{bmatrix}$$

These values are really close to π and Ω respectively.

Question 8 :

For investor Λ with these views, with risk aversion $\bar{\lambda}$ and $c = 10$, we get :

$$w_{\Lambda} = \frac{1}{\bar{\lambda}} \Omega_{BL}^{-1} \tilde{\mu}_{BL}$$

$$w_{\Lambda} = [6.48\%, 40.67\%, 52.06\%]$$

We can compare this result with the initial CWP efficient weights :

$$w_{CWP} = [6.16\%, 41.43\%, 52.42\%]$$

We can conclude that the views have been taken into account, the Johnson n Johnson weight has been reduced by almost 1%, and the Ford weight has increased almost the same % that was lost by the JPMorgan weight. The changes are still relatively small. Everything was invested in the risky assets, risk-free asset remain untouched.

For $c = 1000$, we get :

$$w_{\Lambda} = [26.57\%, -5.16\%, 31.97\%]$$

This time, the risk free asset has nearly half the weight of the portfolio, and the views of the investor have been applied to the excess : Johnson n Johnson is now on a short position, and the other two are on long positions.

We could say that the parameter c is a parameter that controls how much the views of the investor Λ are integrated in the computations.